

Zero-phase dictionary for unequal exponents

Claude, with Daniel Murfet

2026-09-07

Abstract

Unit 198, the first (and, it turns out, only necessary) unit of Astra #20's programme B. The quadratic-kernel chart programme (**twoDGeneral**, parameter n , kernel $e^{-\beta n s^2 + \beta \sqrt{n} s a}$) and the monomial normal-moment programme (**monomialBoxReal**, kernel $e^{-\beta N u^{2k}}$) are tied together at zero phase and *unequal* exponent ratios $p_1 = (h_1 + 1)/k_1 < p_2 = (h_2 + 1)/k_2$: the unit-cutoff zero-phase chart integral *is* the two-dimensional unit-box monomial integral at the same parameter, so the mixed-ratio monomial theorem transports to

$$n^{p_1/2} \int_0^1 \int_0^1 u^{h_1} v^{h_2} e^{-\beta n (u^{k_1} v^{k_2})^2} dv du \longrightarrow \frac{\Gamma(p_1/2)}{2k_1 \beta^{p_1/2} (h_2 + 1 - p_1 k_2)}$$

(**headline_quadratic_zeroPhase_mixed**, plus the swapped version), and uniqueness of limits against the quadratic programme's own no-log theorem evaluates its constant in closed form: $\text{noLogConst}(\beta, 0, p_1, p_2) = \Gamma(p_1/2) \beta^{-p_1/2} / (2(p_2 - p_1))$. Zero **sorry/axiom**; 9075 jobs.

1 The things formalised

```
theorem twoDGeneral_zeroPhase_eq_twoDamp (hn : 0 < n) :
  twoDGeneral 0 b n h h k k = twoDamp b (√n) h h k k (fun _ _ => 1)
theorem twoDGeneral_zeroPhase_eq_monomial (hn : 0 < n) :
  twoDGeneral 0 1 n h h k k = monomialBoxReal 2 ![h, h] ![k, k] n
theorem multCount_two_mixed / monomialMixedConst_two_mixed :
  monomialMixedConst ![h, h] ![k, k] 1 = Γ 1 * ^(-1) / (2k (h + 1 - 2k 1))
theorem monomial_two_mixed_tendsto (hlt : p < p) :
  Tendsto (fun n => monomialBoxReal 2 ![h, h] ![k, k] n / n^(-(p/2))) atTop
  ( (monomialMixedConst ![h, h] ![k, k] (p/2)) )
theorem headline_quadratic_zeroPhase_mixed (hlt : p < p) :
  Tendsto (fun n => n^(p/2) * twoDGeneral 0 1 n h h k k) atTop
  ( (Γ (p/2) / (2k * ^-(p/2) * (h + 1 - p k))) )
theorem headline_quadratic_zeroPhase_mixed' (hgt : p < p) : ... (coordinates exchanged)
theorem noLogConst_zeroPhase_eq_monomialMixedConst (hlt : p < p) :
  1 / (k k) * noLogConst 0 p p = monomialMixedConst ![h, h] ![k, k] (p/2)
theorem noLogConst_zeroPhase_eq (hlt : p < p) :
  noLogConst 0 p p = Γ (p/2) * ^(-(p/2)) / (2 (p - p))
```

2 Mathematics

Dictionary. With $a = 0$ the quadratic kernel is $e^{-\beta n (u^{k_1} v^{k_2})^2} = e^{-\beta n u^{2k_1} v^{2k_2}}$, so the monomial parameter is n itself (not n^2 , contrast unit 193 where **twoDamp** is parametrised by $N = \sqrt{n}$). The proof goes through unit 193's **twoDamp_const_eq_monomialCutoff** at $N = \sqrt{n}$ and $(\sqrt{n})^2 = n$.

Transport. The monomial mixed theorem at $\lambda = p_1/2$ has $J = \{0\}$, $|J| = 1$, so the logarithmic power is $(\log n)^0$ and the constant is $\Gamma(\lambda)\beta^{-\lambda}/0! \cdot \frac{1}{2k_1} \cdot \frac{1}{h_2+1-2k_2\lambda}$. No factor of two arises anywhere (no square in the parameter, no logarithm), which is the audit Astra #20 asked for.

Constant cross-validation. The quadratic programme proves $\text{twoDGeneral} \sim \frac{1}{k_1 k_2} b^{k_2(p_2-p_1)} \text{noLogConst}(\beta, a, p_1, p_2) n^{-p_1/2}$ with $\text{noLogConst}(\beta, a, p_1, p_2) = \int_0^\infty x^{p_1-p_2-1} \int_0^x t^{p_2-1} e^{-\beta t^2 + \beta t a} dt dx$. At $a = 0$, $b = 1$ both programmes describe the same function, so $\frac{1}{k_1 k_2} \text{noLogConst}(\beta, 0, p_1, p_2) = \frac{\Gamma(p_1/2)\beta^{-p_1/2}}{2k_1 k_2(p_2-p_1)}$ (using $h_2 + 1 = p_2 k_2$), i.e. $\text{noLogConst}(\beta, 0, p_1, p_2) = \Gamma(p_1/2)\beta^{-p_1/2}/(2(p_2 - p_1))$. This is exactly what Fubini gives directly ($\int_t^\infty x^{p_1-p_2-1} dx = t^{p_1-p_2}/(p_2 - p_1)$, then $\int_0^\infty t^{p_1-1} e^{-\beta t^2} dt = \frac{1}{2}\beta^{-p_1/2}\Gamma(p_1/2)$), but the Lean proof is by uniqueness of limits and so is valid only for ratios realised by natural data. Numerical check at $\beta = 1.3$, $(h, k) = ((0, 2), (1, 1))$, $p_1 = 1$, $p_2 = 3$: quadrature 0.38863621594708 vs closed form 0.38863621594708; $n^{1/2}Q_n$ at $n = 10^6$ is 0.388636.

3 Lean notes

- `ring` cannot prove $\frac{X}{2k_1(h_2+1-2k_2l)} = X \cdot \frac{1}{2k_1} \cdot \frac{1}{h_2+1-2k_2l}$ because it expands the product inside the inverse into a sum first and then treats the inverse of the sum as an atom. `simp only [div_eq_mul_inv, mul_inv, one_mul]` before `ring` distributes the inverse over the product syntactically.
- Rewriting a goal $(h+1)/(2k) \quad p/2$ with $\leftarrow hp$ fails (no p_2 present); establish the equality `ratioExp ... 1 = p/2` first and rewrite with it.
- `monomialBoxReal_mixed_tendsto 1 ! [h, h] ! [k, k]` elaborates without fuss (`Fin (1+1)` vs `Fin 2`).

4 Status

Astra #20 programme B is closed in one unit (the general mixed- d transport it mentioned as a possible follow-on is already available in the monomial programme; the quadratic side has no arbitrary-dimensional mixed theorem beyond one dominated coordinate, and nothing in the paper requires one at zero phase). Next: gate for programme A (general- d random-phase leading term).